RemoVia

Setup & TV Pairing Guide - beginner edition

RemoVia is one remote on your phone for every smart TV in the house - Roku, Samsung, Google TV, Android TV and Fire TV. This guide is written for beginners: it tells you exactly what to press, what you will see when it works, and what to do if it doesn't. Nothing here can break your TV or your computer. The bridge runs on a Mac or a Windows PC - wherever the two differ, the guide labels the Mac path and the Windows path, and you follow only yours. The same guide is on the web at iecincglobal.com/removia-guide.html.

Before you start - five words we will use

Bridge: a small program that runs on your computer - a Mac or a Windows PC - and talks to your TVs for you. Your phone tells the bridge; the bridge tells the TV.

Terminal: a window where you type commands. On a Mac it is called Terminal; on Windows, Command Prompt. Don't worry - we tell you exactly what to type, word for word.

Token: a long secret password that your bridge makes for you. It proves the phone asking is yours.

IP address: your TV's number on your home Wi-Fi - like a house number on a street. We show you where the TV displays its own number.

Tailscale: a free app that builds a private, safe connection between your phone and your computer, so the remote works from anywhere.

Part 1 - What you need

- A computer - a Mac or a Windows PC - on the same Wi-Fi as your TVs. The bridge runs on it. (It needs Python 3.11 or newer. Most Macs are ready; on Windows, Part 2W installs Python with you, step by step. Either way, the bridge sets up everything else by itself the first time it runs.)
- Your phone. The remote is one web page - nothing to install from an app store for the remote itself.
- The RemoVia bridge ZIP file (the download). Unzipping it makes a folder called removia-main.
- A free Tailscale account. You create it by signing in during Part 3 - no separate sign-up trip.
- Each TV's own remote control. Pairing is approved on the TV screen, once per TV.

TVs RemoVia works with: Roku - Samsung - Google TV and Android TV sets (Sony, TCL, Hisense, Philips, Xiaomi, NVIDIA Shield) - Amazon Fire TV. LG (webOS) sets are found and saved, but cannot be controlled yet.

CHECKPOINT: You should now have: the unzipped removia-main folder in your Downloads, your phone in hand, each TV's own remote nearby.

Part 2 - Get the bridge running on your computer

There are two paths through this part - one for a Mac, one for a Windows PC. Follow only yours. Both end at the same place: the bridge running, and your token saved somewhere safe.

2M - On a Mac

Mac owners type a handful of short lines into an app called Terminal. Terminal sounds technical; it is just a window where you type. We give you every line exactly.

1. Press: click the magnifying glass at the top-right of the screen (that's called Spotlight). Type the word Terminal. Press Enter.

   YOU SHOULD SEE: A plain window opens with a few lines of text and a blinking cursor. That window is Term did it.

2. Know this: you can copy and paste every command in this guide instead of typing - copy with Cmd+C, paste into Terminal with Cmd+V, then press Enter to run it. And nothing you type here can break your TV or your computer - the worst that can happen is an error message, and you just try again.

3. Type this exactly, then press Enter:

   ```
   cd ~/Downloads/removia-main
   ```

   YOU SHOULD SEE: The text at the left of the window now ends with removia-main. That means you are folder.

   IF NOT: If it says 'no such file or folder', the ZIP was not unzipped yet. Go to Downloads, double-click the ZIP, then try the line again.

4. Type this exactly, then press Enter:

   ```
   ./start.sh
   ```

   YOU SHOULD SEE: The first time, several minutes of text scroll by while the bridge builds itself - that is no finished when a line says 'Uvicorn running on http://0.0.0.0:8765'. That line means: the bric

   IF NOT: The Mac may pop up a question - 'allow Python to accept incoming connections?' Click Allow. If the text ends with 'address already in use', an old copy of the bridge is already running - see the Troubleshooting table in Part 7.

5. Your token appears once. In that first run of text, look for the words:

   ```
   First run: your RemoVia bridge access token is:
   ```

Under those words is a long line of letters and numbers. That is your token. Your bridge made it, on your computer, for you - nobody else has it, and IEC cannot see it. Do this: select that long line, copy it (Cmd+C), and paste it somewhere safe, like your Notes app. You will paste it into the remote page in Part 3.

IF NOT: If you lose it: the token also lives in a settings file on your Mac called config.json, inside a folder named .tv-remote in your home folder. Open that file only to read your token. Never send that file, or the token, to anyone - not even to us. Whoever has the token and a path to your bridge can control your TVs.

6. Leave this Terminal window open. The bridge runs while the window stays open. Closing the window turns the remote off - that is normal, not broken. To turn it back on, open Terminal and run the same two lines from steps 3 and 4 again.

2W - On a Windows PC

Windows owners type no commands at all in this part. You install one program (Python), then start the bridge by double-clicking a file.

1. Press: open your web browser and go to python.org/downloads. Click the big Download Python button, then open the file it downloads. On the very first setup screen: tick the box at the bottom

labelled 'Add python.exe to PATH' - this tick matters - then click Install Now.

YOU SHOULD SEE: 'Setup was successful'. Click Close.

IF NOT: If you clicked Install without the tick, don't worry: open Settings > Apps, uninstall Python, then run the installer again and tick the box this time. Without the tick, the bridge cannot find Python and the next steps do nothing.

2. Press: find the downloaded RemoVia ZIP (in your Downloads folder). Right-click it > Extract All > Extract. Then open your user folder: open File Explorer and go to This PC > your hard drive (C:) > Users > your name. Move the unzipped removia-main folder into your user folder, and rename it to tv-remote.

   YOU SHOULD SEE: A folder named tv-remote sitting in your user folder, and inside it a file named Start-RemoV

   IF NOT: If double-clicking the ZIP just opens it like a folder, you have not extracted it yet - go back and use Right-click > Extract All. The bridge cannot run from inside the ZIP.

3. Press: open the tv-remote folder and double-click Start-RemoVia.bat.

   YOU SHOULD SEE: A black window opens. The first time, several minutes of text scroll by while the bridge bu that is normal. It is finished when a line says 'Uvicorn running on http://0.0.0.0:8765'. means: the bridge is ON.

   IF NOT: If the window closes instantly or says Python was not found, the tick in step 1 was missed - redo step 1 (uninstall, reinstall, tick the box). If the text ends with 'address already in use', an old copy of the bridge is still running - see the Troubleshooting table in Part 7.

4. The firewall question. Windows may pop up a message: 'Windows Defender Firewall has blocked some features of Python'. Press: tick Private networks, then click Allow access.

   YOU SHOULD SEE: The message closes and nothing else changes. Without this Allow, your phone cannot bridge.

   IF NOT: If no message appears, that is fine too - it may wait until your phone first connects. Give it the same answer then: Private networks > Allow access.

5. Your token appears once. In that first run of text in the black window, look for the words:

```
First run: your RemoVia bridge access token is:
```

Under those words is a long line of letters and numbers. That is your token. Your bridge made it, on your computer, for you - nobody else has it, and IEC cannot see it. Do this: select that long line with the mouse, copy it (Ctrl+C), and paste it somewhere safe, like a note on your phone. You will paste it into the remote page in Part 3.

IF NOT: If you lose it: the token also lives in a settings file called config.json, inside a folder named .tv-remote in your user folder (in full: C:\Users\your name\.tv-remote\config.json). Open that file only to read your token. Never send that file, or the token, to anyone - not even to us.

6. Leave the black window open. The bridge runs while the window stays open. Closing the window turns the remote off - that is normal, not broken. To turn it back on, double-click Start-RemoVia.bat again.

7. Nice extra - start the bridge automatically: press the Windows key + R together. A small 'Run' box opens. Type shell:startup and press Enter - a folder opens. In your tv-remote folder, right-click Start-RemoVia.bat > Copy. Back in the startup folder, right-click an empty spot > Paste shortcut.

   YOU SHOULD SEE: A shortcut to Start-RemoVia in the startup folder. From now on, the bridge starts by itself e you sign in to Windows.

CHECKPOINT: You should now see: 'Uvicorn running' in the bridge window (Terminal on a Mac, the black windo
Windows), the window still open, and your token saved somewhere safe. Windows owners: from her
you can ignore the word 'Terminal' - the only typing left is one line in Part 3D, and it is labelle
Windows too.

Part 3 - Connect your phone securely (Tailscale)

Why this part exists, in one sentence: the remote page (removia.iecincglobal.com) is a secure web page, and browsers will not let a secure page talk directly to the bridge in your house. Tailscale is the private path between them - think of it as a locked corridor that only your devices can enter. It is free.

3A - On your phone

1. Press: Open the App Store. Search for Tailscale. Tap Get, then Open.
2. Press: Sign in. Use the same account you will use on the computer (for example, the same Google account). The account is created the first time you sign in.

   YOU SHOULD SEE: Your phone warns you that Tailscale wants to add a VPN configuration. This is expe
   private corridor is what the warning is about. Tap Allow, and type your phone passcode if i

   YOU SHOULD SEE: The Tailscale app says Connected and lists your phone.

3B - On your computer (Mac or Windows)

1. Press: Install Tailscale on the computer as well (from tailscale.com/download - on a Mac you can also use the Mac App Store; on Windows, run the installer you download), open it, and sign in with the same account as the phone.

   YOU SHOULD SEE: A short welcome survey may appear ('Help us better understand your product needs'). Fo
   your primary reason for using Tailscale?' pick Personal or At-Home Use, pick any answe
   role, and press Submit. It changes nothing about your setup.

   YOU SHOULD SEE: The app says Connected. Your computer and your phone now both appear in the device li

3C - Turn on HTTPS (one time only)

1. Press: In your web browser on the computer, go to console.tailscale.com/admin/dns. Sign in with the same account if it asks.
2. Press: Scroll to HTTPS Certificates and click Enable HTTPS.

   YOU SHOULD SEE: The section now shows HTTPS as enabled. You never need this page again.

3D - Give the bridge its secure address

1. Press: On a Mac, open a new Terminal window (in Terminal's Window menu, or Cmd+N) - leave the bridge window from Part 2 untouched. On Windows, press the Start button, type cmd, and press Enter - a Command Prompt window opens (leave the black bridge window from Part 2 untouched). On either one, type this exactly, then press Enter:

   ```
   tailscale serve --bg 8765
   ```

   YOU SHOULD SEE: The words 'Serve started and running in the background', and an address that
   https://your-computer.tail1234.ts.net. Write this address down - it is your bridge's pri
   address.

   IF NOT: Hiccup 1: the command may stop and say 'Serve is not enabled on your tailnet. To enable, visit:'

followed by a link. That is the one-time switch from Part 3C asking to be turned on. On a Mac, hold Cmd and click the link in Terminal. On Windows, copy the link and paste it into your browser. Sign in, click Enable - then type the same command again and press Enter.

IF NOT: Hiccup 2: a message saying 'Another client is changing the serve config' or 'etag mismatch - please try again' simply means two changes bumped into each other. Run the same command once more. It works.

3E - Tell the remote page where your bridge lives

1. Press: On your phone, open removia.iecincglobal.com. Tap the gear (Settings) at the top-right.
2. Press: In Bridge address, paste the ts.net address from 3D. In Token, paste your bridge token from Part 2. Tap Save. Then tap Test.

   YOU SHOULD SEE: The dot at the top of the page turns green and the word Connected appears. Your re works from anywhere - home or away - and nothing in your house is exposed to the public

CHECKPOINT: You should now see: a green dot and 'Connected' on the RemoVia page on your phone, and both phone and your computer listed as Connected in Tailscale.

Part 4 - Add your TV, and find its number (IP address)

1. Press: In RemoVia on your phone, tap + TV, then tap Scan the network.

   YOU SHOULD SEE: After a few seconds, your TV appears in a list. Tap Add next to it, type a room name (fo BedRoom), and save.

   IF NOT: If the scan finds nothing, the TV may be asleep or on a different Wi-Fi. Wake it, check it is on the same Wi-Fi as your computer, and scan again. Still nothing? Add it by hand - the next steps show how.
2. To add by hand, you need the TV's IP address - its house number on your Wi-Fi. The TV displays it in its own settings. Samsung: with the TV's own remote, press Settings > General > Network > Network Status > IP Settings. You will see a screen listing IP Address, Subnet Mask and Gateway. Write down the IP Address line. Then press Cancel to leave without changing anything. Roku: with the TV's own remote, press Settings > Network > About. Write down the IP address shown there.
3. Press: Back in RemoVia's + TV screen, pick the TV's brand, type the IP address you wrote down, type a room name, and tap Add this TV.

The one mistake almost everyone makes: entering the Gateway number instead of the IP Address. The Gateway home networks it is 10.0.0.1. The two lines sit next to each other on gateway, the bridge knocks on the router's door and pairing fails with use the line labelled IP Address.

CHECKPOINT: You should now see: your TV as a named button (for example 'BedRoom') at the top of the Rem page.

Part 5 - Pair by brand

Pairing means the TV agrees to listen to RemoVia. It happens once per TV - after that, the TV remembers and you never do this again.

Roku - no pairing, one setting on the TV

There is nothing to pair in the app. But the Roku must be allowed to listen. This is done on the TV, with the TV's own remote - not in RemoVia:

1. Press (TV remote): Settings > System > Advanced system settings > Control by mobile apps > Network access.
2. Press: choose Permissive.

   YOU SHOULD SEE: Text under it saying control by mobile apps 'is enabled always, but accepts commands private network addresses or the same subnet'. That sentence is the good one - it mean your home network may control the TV, and nothing outside can.

   IF NOT: If Network access is left on Limited, every button in RemoVia answers with error (403).

CHECKPOINT: Test it: in RemoVia, select the Roku and press Home. The TV should jump to its home screen.

Samsung - press Allow on the TV

Read this first: the pairing prompt appears on the TV, not on your phone - and it only stays on screen a short time. I the TV's own remote in your hand before you tap Start pairing. The TV must be fully on, showi normal channel or app, with all menus closed.

1. Press (phone): in RemoVia, select your Samsung TV, then tap Start pairing.

   YOU SHOULD SEE: On the TV: a box pops up saying 'RemoVia - Your permission is required to connect your TV.' The top choice is Allow.
2. Press (TV remote): press the middle / OK button to choose Allow.
3. Press (phone): tap Done.
4. Test it: in RemoVia, press VOL +. You should see: the volume bar moves on the TV. Pairing is done - forever.

   IF NOT: If the phone says the TV 'did not accept in time', the prompt timed out or a menu was open. Close all menus on the TV and repeat from step 1, remote in hand. If no prompt ever appears: on the TV, open Settings > General > External Device Manager > Device Connect Manager and set Access Notification to Always On, then try again. (Also under Settings > General > Network > Expert Settings, 'Power with Mobile' should be on. Menu names vary a little by model.)

CHECKPOINT: You should now see: the Samsung's volume move when you press VOL + in RemoVia.

Google TV / Android TV - type the PIN the TV shows you

For Sony, TCL, Hisense, Philips, Xiaomi and NVIDIA Shield sets:

1. Press (phone): select the TV and tap Start pairing.

   YOU SHOULD SEE: On the TV: a 6-digit PIN - six numbers on the TV screen.
2. Press (phone): type those six numbers into RemoVia and tap Pair with this PIN.

CHECKPOINT: Test it: press any button in RemoVia - the TV responds.

Fire TV - one TV setting, then Allow

1. Press (TV remote): Settings > My Fire TV > Developer Options > ADB Debugging > On. (This simply lets the TV accept a remote connection.)
2. Press (phone): select the TV and tap Start pairing.

   YOU SHOULD SEE: On the TV: a prompt asking to allow the connection. Tick 'Always allow' if it is shown, the

Allow.

3. Press (phone): tap Done.

CHECKPOINT: Test it: press any button in RemoVia - the TV responds.

LG (webOS)

RemoVia can find an LG TV and save it, but it cannot control it yet - there is no pairing step. If you have an LG, skip it for now; the other brands above are fully supported.

Part 6 - Make it yours: remote faces

The remote has three looks, called faces. Changing the face changes only how the remote looks - every button does exactly the same thing as before.

1. Press: on the RemoVia page on your phone, tap the gear (Settings) at the top-right.
2. Press: find Remote face and tap the look you want. Midnight - the original dark look, easy on the eyes at night. Daylight - a light look, easier to read in bright rooms and sunshine. High contrast - black, white and yellow, as bold as it gets; best if small text is hard to read.

   YOU SHOULD SEE: The whole remote changes look the moment you tap - background, buttons and text tog
   just one piece.
3. Know this: your choice is saved on your phone. Close the page and come back tomorrow - your face is still there. Every phone in the house can pick its own.

CHECKPOINT: You should now see: the remote wearing the face you picked - and it is still wearing it after you close
reopen the page.

Part 7 - Troubleshooting

Find the words you see in bold. Every fix is short, and none of them can harm anything.

Error (403) - Roku

Fix: On the TV: Settings > System > Advanced system settings > Control by mobile apps > Network access = Permissive. (Part 5, Roku.) The Roku's Network access is set to Limited, so it refuses app commands.

ConnectionRefusedError

Fix: The bridge used the wrong IP (often the gateway, 10.0.0.1), or the TV is off. Read the TV's real IP Address in its Network settings (Part 4), re-add the TV with that address, and make sure the TV is on.

"The TV did not accept in time, or it is off"

Fix: The pairing prompt on the TV timed out before anyone pressed Allow. TV fully on, all menus closed, TV remote in hand. Tap Start pairing and press Allow on the TV promptly.

"Stopped answering"

Fix: The connection between the bridge and the TV dropped. Restart the bridge. Mac: click its Terminal

window, press Ctrl+C, then type ./start.sh and press Enter. Windows: close the black window, then double-click Start-RemoVia.bat again.

"Address already in use" when starting the bridge

Fix: The old bridge is still running. Stop it first. Mac: press Ctrl+C in its Terminal window - or, in a new Terminal window, type: kill $(lsof -ti :8765) and press Enter - then run ./start.sh again. Windows: close the old black window, then double-click Start-RemoVia.bat.

"No such file or folder" after the cd line (Mac)

Fix: The bridge folder is not where Terminal is looking - usually the ZIP was never unzipped. In Downloads, double-click the ZIP so the removia-main folder appears. Then run the cd line again.

Red dot - the page cannot reach the bridge

Fix: Tailscale or the bridge is not running, or the address/token in Settings is wrong. Check Tailscale says Connected on both phone and computer, with the same account. Check the bridge window (Terminal on Mac, the black window on Windows) is open and says 'Uvicorn running'. Check the Bridge address and Token in the page's Settings match Part 3 exactly.

Windows: the black window says Python was not found

Fix: Python was installed without the 'Add python.exe to PATH' tick, so the bridge cannot find it. Uninstall Python (Settings > Apps), install it again, and tick Add python.exe to PATH on the first setup screen (Part 2W, step 1).

Phone cannot connect at home

Fix: Your computer's firewall may be blocking the bridge. Mac: System Settings > Network > Firewall > Options > allow incoming connections for Python. Windows: you may have missed the firewall question in Part 2W - open Windows Security > Firewall & network protection > Allow an app through firewall, find Python, tick Private, and click OK.

Still stuck? Go back to the last checkpoint you passed and re-read that part's 'You should see' line. Almost every problem is one of the rows above.

Part 8 - Safety notes

- Your token is a key to your TVs. Your bridge made it, on your computer. Never post it, share it, or share the config.json file that holds it (in the .tv-remote folder in your home folder - on Windows, C:\Users\your name\.tv-remote\config.json) - with anyone, including us.
- Tailscale keeps everything private. Only devices signed in to your account can reach your bridge. Nothing in your house is exposed to the public internet, and no router port-forwarding is needed - never set up port-forwarding for the bridge.
- Pairing prompts appear on the TV itself. Press Allow only for a prompt you just started. The prompt names the app - RemoVia.
- The bridge window is the on/off switch. Bridge window open = remote on. Window closed = remote off. That is normal, and reopening it (Part 2) turns everything back on - your TVs stay paired.

Prefer paper? This PDF is the full guide - the same steps as the web page at iecincglobal.com/removia-guide.html, ready to print or share with anyone in the house.